

DAC of CO₂ via NaOH Absorption & Electrochemical Regeneration

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This work investigates a direct air capture process based on CO₂ absorption in aqueous NaOH and EC regeneration of spent solution. A python based model was developed for both the packed absorption column and the electrochemical regeneration unit in order to evaluate mass transfer, reaction behavior, material balances, and energy demand. Under the selected base case conditions, the absorber reduced the CO₂ concentration in air from 420 ppm to 1.43 ppm, corresponding to a capture efficiency of 99.6%. The EC cell recovered 96.5% of the captured CO₂ and the regenerated 1.72 M NaOH product with 92% purity at a specific energy demand of 160 kJ mol⁻¹ CO₂. The results show the EC regeneration is a promising low-temperature for alkaline DAC systems.

1. Introduction

Anthropogenic CO₂ emissions have increased the atmospheric CO₂ concentration from about 280 ppm in the pre-industrial era to more than 420 ppm today. Carbon dioxide removal must complement direct emission reduction, and direct air capture (DAC) is one of the few scalable options that removes CO₂ directly from ambient air independent of point sources. [1,2]

Current DAC technologies are mainly based on either liquid alkaline solvents or solid sorbents. Among these, alkaline systems using NaOH or KOH are attractive because of their strong affinity for CO₂ and well-established reaction chemistry. However, their practical deployment is limited by the regeneration step, which in conventional processes typically relies on energy-intensive thermal treatment, such as CaCO₃ calcination at high temperature. [1,3] Electrochemical regeneration has emerged as a promising alternative because it can recover alkali solution and release a concentrated CO₂ stream at comparatively mild conditions, potentially reducing thermal demand and simplifying process integration.

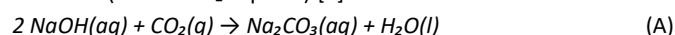
The objective of this work is to develop and evaluate a coupled DAC process based on CO₂ absorption in aqueous NaOH and electrochemical regeneration of the carbonate-rich solution. For this purpose, a Python-based model was developed for the packed absorber, and a dynamic model was formulated for the electrochemical regeneration unit. In addition, a surrogate loading model was used to connect the single-pass absorber behavior with the cyclic operation of the electrochemical cell. The integrated framework was then used to establish mass and energy balances, analyze process performance, and identify key operating trade-offs for the proposed DAC system.

Process Overview

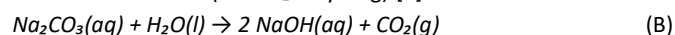
Ambient air (≈ 420 ppm CO₂) is contacted with a recirculating 2.0 M NaOH solution in a structured packing column. The column is about 0.29 m in diameter and 15 m packed height, with a gas flow of 35 m³/h and a liquid circulation of 40 L/min, corresponding to a liquid residence time of roughly 5 min per pass. Each pass removes $\approx 99.7\%$ of CO₂ from the gas, but only loads the liquid with about 2.5×10^{-4} mol/L of DIC (CO₃²⁻-based), leaving the NaOH concentration essentially unchanged at ≈ 2.0 M.

Over many hours of operation, the NaOH solution is meant to transition from mostly NaOH to mostly Na₂CO₃, and this carbonate-rich solution becomes the feed to the electrochemical cell, which regenerates NaOH and releases purified CO₂. The EC stack in the design case consists of 7 cells of 500 cm² each, operated at ≈ 92 A/m² (≈ 4.6 A stack current, ≈ 5 V stack voltage, ≈ 23 W DC power), and runs until the “acid” tank (Na₂CO₃-rich side) reaches pH 5. Detailed summary of Input and output streams of absorber and EC cell is in the jupyter notebook files provided with the report. Block flow diagram showing how the overall process looks like is given in the appendix.

Absorber (overall CO₂ capture):[4]



Electrochemical cell (with H₂ recycling):[4]



Surrogate absorber model as the bridge

The detailed absorber model provides accurate single-pass behaviour, but it is computationally expensive to run over thousands of passes. To bridge between the absorber and the EC cell, the surrogate absorber model is introduced:

- Treats the absorber loop as a 200 L perfectly mixed tank recirculating at 40 L/min.
- Uses the detailed absorber result ($\Delta\text{C_DIC_ref} \approx 2.5 \times 10^{-4}$ mol/L per pass at 2.0 M NaOH) as a reference per-pass DIC increment.

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- Scales this increment with NaOH concentration using the Hatta number ($Ha \propto \sqrt{c_{OH}}$) and an enhancement factor, preserving the transition from strongly reaction-enhanced to weaker absorption as NaOH is consumed.

On each “pass” (≈ 5 min), the surrogate decreases the NaOH concentration and increases the Na_2CO_3 concentration by a ΔC_{DIC} that depends on the current NaOH level. The model stops when NaOH has dropped from 2.0 M to ≈ 0.10 M and Na_2CO_3 has increased to ≈ 0.95 M, which occurs after about 3970 passes, or ≈ 331 hours (≈ 14 days) of operation. At that point, the absorber loop has captured ≈ 190 mol CO_2 (≈ 8.4 kg) in the 200 L solution, and this final composition (≈ 0.95 M Na_2CO_3 + 0.10 M NaOH in ≈ 200 L) is taken as the feed to the EC model.

2. Mathematical Model Absorber

The absorber was modeled as a one-dimensional, steady-state, counter-current packed column for direct air capture of CO_2 using aqueous NaOH as the reactive solvent. The axial coordinate z was defined from the bottom to the top of the column, such that $z=0$ corresponds to the gas inlet and liquid outlet, whereas $z=Z$ corresponds to the gas outlet and liquid inlet. In the implementation, the isothermal formulation is represented by the Absorber class, while the non-isothermal formulation is represented by the AdiabaticAbsorber class.

The gas phase contains dilute CO_2 in humid air, whereas the liquid phase consists of aqueous NaOH together with dissolved carbonate species formed during absorption. The model predicts axial profiles of gas-phase CO_2 , water vapour, dissolved inorganic carbon concentration, pH, free NaOH concentration, enhancement factor, overall mass-transfer coefficient, and, for the adiabatic case, gas and liquid temperatures.

Absorber Model assumptions

The model assumes steady-state operation and one-dimensional axial transport through the packed section. Axial dispersion is neglected in the solved governing equations, and the total pressure is fixed at atmospheric pressure throughout the column. The carbonate system in the liquid phase is assumed to be in local equilibrium at each axial position, whereas the interphase transfer of CO_2 is described by a two-film model with chemical enhancement due to reaction with hydroxide ions.

For the isothermal model, the gas and liquid temperatures are fixed at their inlet values over the full column height. For the adiabatic model, additional coupled energy balances are solved for the gas and liquid phases.

Governing mass balances

The isothermal model solves balances for gas-phase CO_2 , liquid-phase dissolved inorganic carbon, and gas-phase water vapour.

These equations are written in terms of the gas-phase mole ratios Y_{CO_2} and Y_{H_2O} , and the liquid-phase dissolved inorganic carbon concentration C_{DIC} . The axial balances are given by Eqs. (1)–(3). [5,6]

$$\frac{dY_{CO_2}}{dz} = -\frac{a_w N_{CO_2}}{G_{inert}} \quad (1)$$

$$\frac{dC_{DIC}}{dz} = -\frac{a_w N_{CO_2}}{u_{L,loc} 1000} \quad (2)$$

$$\frac{dY_{H_2O}}{dz} = \frac{a_w N_{H_2O}}{G_{inert}} \quad (1)$$

In Eq. (1), the decrease in gas-phase CO_2 is governed by the local interfacial absorption flux N_{CO_2} , scaled with the wetted interfacial area a_w and the inert gas flow basis G_{inert} . Eq. (2) describes the corresponding increase in dissolved inorganic carbon in the liquid phase, while Eq. (3) accounts for the rise in gas-phase water content due to evaporation from the alkaline solution. The liquid superficial velocity used in Eq. (2) is adjusted locally in the code as a function of DIC loading and is therefore denoted by $u_{L,loc}$.

Governing energy balances

For the adiabatic model, the gas and liquid temperatures are obtained from coupled axial energy balances. The gas-phase energy balance includes interphase sensible heat transfer, whereas the liquid-phase energy balance includes sensible heat exchange, heat release due to absorption and reaction, and latent cooling associated with water evaporation. The governing energy balances are given by Eqs. (4) and (5). [5,6]

$$\frac{dT_G}{dz} = -\frac{a_w Q_{GL}}{G_{mol,loc} C_p, G} \quad (4)$$

$$\frac{dT_L}{dz} = -\frac{a_w (Q_{GL} + Q_{rxn} - Q_{evap})}{u_{L,loc} 1000 C_p, L} \quad (5)$$

According to Eq. (4), the gas temperature changes due to sensible heat exchange between the gas and liquid phases. In Eq. (5), the liquid temperature is influenced by interphase heat transfer, the heat released by CO_2 absorption and reaction, and the latent heat consumed during water evaporation.

CO_2 transfer and kinetics

The local absorption rate of CO_2 is calculated from an overall gas-side driving-force expression. The flux relation used in the model is given by Eq. (6). [7]

$$N_{CO_2} = K_G (Y_{CO_2} - y^*_{CO_2}) \quad (6)$$

Here, K_G is the overall gas-side mass-transfer coefficient and $y^*_{CO_2}$ is the equilibrium interfacial gas-phase mole fraction corresponding to the local dissolved CO_2 concentration. The overall mass-transfer coefficient is obtained from gas-side and liquid-side resistances in series, as shown in Eq. (7). [7]

$$\frac{1}{K_G} = \frac{1}{k_G} + \frac{1}{H_{eff} 1000 E k_L} \quad (7)$$

The model accounts for the chemical reaction between dissolved CO_2 and hydroxide ions by means of an apparent pseudo-first-order rate

constant, given by Eq. (8). The intrinsic reaction constant k_{OH} follows an Arrhenius-type relation, as shown in Eq. (9). [8],[39]

$$k_{app} = k_{OH} f_{ion} c_{OH} - \quad (8)$$

$$k_{OH} = 5.985 \times 10^{13} \exp\left(-\frac{55400}{RT}\right) \quad (9)$$

Using this apparent rate constant, the Hatta number is evaluated according to Eq. (10). [9]

$$Ha = \frac{\sqrt{(k_{app} D_A)}}{k_L} \quad (10)$$

The enhancement factor E is then obtained from the Hatta-number-based formulation given by Eqs. (11) and (12). [9]

$$q = \frac{Ha^2}{2(E_{\infty} - 1)} \quad (11)$$

$$E = -q + \sqrt{\left(q^2 + \frac{E_{\infty} Ha^2}{E_{\infty} - 1} + 1\right)} \quad (12)$$

Equation (10) expresses the competition between reaction and diffusion in the liquid film, while Eqs. (11)-(12) allow the model to transition between weakly enhanced and fast chemically enhanced absorption regimes.

Auxiliary Relations

Thermodynamic, transport, and hydraulic quantities are evaluated locally from the state of the system, and the corresponding correlations are listed in the Appendix. These auxiliary relations are used to determine gas-phase composition, carbonate speciation and CO_2 equilibrium in the liquid phase, as well as the local hydroxide concentration, free NaOH concentration, and enhancement factor. In addition, the model includes correlations for mass transfer, diffusivities, heat capacities, hydraulic and wetting behaviour, water evaporation, and the pH-dependent heat of reaction.

Numerical Solution

The governing equations were implemented in Python using SciPy solvers. For the isothermal model, the system of ordinary differential equations was integrated with `solve_ivp` using the RK45 method, while the unknown bottom value of C_{DIC} was determined using a shooting method combined with the `brentq` root finder. For the adiabatic model, an isothermal warm start was first generated, after which a two-variable shooting procedure with `fsolve` was applied to determine the bottom dissolved inorganic carbon concentration and the bottom liquid temperature. The axial domain is discretized into 401 points for profile evaluation and post-processing.

3. Mathematical Model Electrochemical Regeneration

The electrochemical regeneration unit was modeled as a dynamic, two-reservoir electrochemical system consisting of an acid tank, a base tank, electrodes, and a cation-exchange membrane (CEM). The model tracks transient species inventories in both tanks, membrane transport of sodium and protons, hydroxide generation in the base compartment, CO_2 degassing from the acid compartment, and the temperature evolution of the complete electrochemical unit.

In the implementation, the electrochemical subsystem is represented by the `ElectrochemicalCell` class, while the overall absorber-electrolyzer cycle is coupled through the DAC system routine. The acid tank is initialized with the carbonate-rich solution leaving the absorber, whereas the base tank is seeded with a dilute NaOH solution that is progressively enriched during electrolysis.

The model assumes perfectly mixed acid and base tanks, transient operation, uniform current density over the active membrane area, constant membrane thickness, and one lumped system temperature. Carbonate speciation in the acid tank is assumed to be at local equilibrium at each time step, whereas membrane transport is calculated from a reduced Nernst-Planck formulation including diffusion, migration, Donnan partitioning, and electro-osmotic water drag.

State Variables and Governing Equations

The state vector is written in terms of moles rather than concentrations, which allows the model to account for tank-volume changes caused by water transport and electrochemical consumption. The state vector used in the numerical solution is

$$y = [n_{DIC}, n_{Na,a}, n_{Na,b}, n_{OH,b}, n_{CO_2,g}, n_{res}, T, n_{H_2O,a}, n_{H_2O,b}] \quad (13)$$

Here, n_{DIC} is the total dissolved inorganic carbon in the acid tank, $n_{Na,a}$ and $n_{Na,b}$ are the sodium inventories in the acid and base tanks, $n_{OH,b}$ is the hydroxide inventory in the base tank, $n_{CO_2,g}$ is the cumulative amount of released carbon dioxide, n_{res} is the residual NaOH inventory in the acid tank, and T is the lumped system temperature.

Because the code solves mole balances, the liquid volumes are recalculated at every time step from partial molar volumes. The concentrations used in speciation, conductivity, and membrane transport are therefore obtained from Eqs. (14)–(17).

$$V_a = n_{H_2O,a} \bar{V}_{H_2O} + n_{Na,a} \bar{V}_{Na^+} + n_{DIC} \bar{V}_{DIC} \quad (14)$$

$$V_b = n_{H_2O,b} \bar{V}_{H_2O} + n_{Na,b} \bar{V}_{Na^+} + n_{OH,b} \bar{V}_{OH^-} \quad (15)$$

$$c_{i,a} = \frac{n_{i,a}}{V_a} \quad (16)$$

$$c_{i,b} = \frac{n_{i,b}}{V_b} \quad (17)$$

The electrochemical regeneration rate is linked to the applied current density through Faraday's law and an empirical Faradaic-efficiency correction. The total current and current efficiency are calculated as [23]

$$I_{tot} = j_{applied} A_{cell} N_{cell} \quad (18)$$

$$\eta_F = \eta_{F,max} \max\left(1 - \frac{j_{parasitic}}{|j_{applied}|}, 0\right) \quad (19)$$

The corresponding molar electron-transfer rate is then written as [23]

$$r_e = \eta_F \frac{I_{tot}}{F} \quad (20)$$

The transient component balances solved by the model are

$$\frac{dn_{DIC}}{dt} = -R_{degas} \quad (21)$$

$$\frac{dn_{Na,a}}{dt} = -|R_{Na}| \quad (22)$$

$$\frac{dn_{Na,b}}{dt} = |R_{Na}| \quad (23)$$

$$\frac{dn_{OH,b}}{dt} = r_e - |R_{H,back}| \quad (24)$$

$$\frac{dn_{CO_2,g}}{dt} = R_{degas} \quad (25)$$

$$\frac{dn_{res}}{dt} = -R_{neutralize} \quad (26)$$

$$\frac{dn_{H_2O,a}}{dt} = -|R_{H_2O,drag}| \quad (27)$$

$$\frac{dn_{H_2O,b}}{dt} = -r_e + |R_{H_2O,drag}| + |R_{H,back}| \quad (28)$$

In the code, the membrane fluxes are first computed on an area basis and then multiplied by the total active area $A_{tot} = A_{cell}N_{cell}$ to obtain the molar rates between compartments. The acid-side CO_2 release is modeled by a gas-liquid stripping relation, and the residual NaOH consumption is smoothed with a logistic switching function so that proton neutralization becomes important only while free caustic remains in the acid tank.

The reversible voltage of the cell is written from the pH difference between the base and acid tanks, while the total cell voltage includes activation, ohmic, membrane, and concentration overpotentials: [24,25]

$$E_{rev} = \frac{RT}{F} 2.303 (pH_b - pH_a) \quad (29)$$

$$E_{cell} = E_{rev,act} + \eta_{an} + \eta_{cat} + \eta_{ohm} + |\eta_{mem}| + \eta_{conc,an} + \eta_{conc,cat} \quad (30)$$

The electrical power is then obtained from $P = E_{cell}I_{tot}$, and the transient energy demand is calculated by time integration of the power profile. The simulation is terminated when the acid-tank pH reaches the specified stopping criterion.

A lumped thermal balance is also included in the electrochemical model. Irreversible electrical losses, reversible entropic heating or cooling, and external heat exchange with the surroundings are combined in the temperature balance. [34,35]

$$\frac{dT}{dt} = \frac{I_{tot}\eta_{irr} + I_{tot}T\frac{\Delta S}{F} - UA_{ext}(T - T_{amb})}{m_a C_{p,a} + m_b C_{p,b}} \quad (31)$$

Auxiliary Relations

The constitutive relations required for carbonate speciation, activity-corrected pH, ionic diffusivities, ionic conductivities, electrolyte conductivity, membrane transport, electrode kinetics, degassing, and thermal effects are summarized in Appendix B. These correlations provide the local transport and thermodynamic properties required at each time step of the transient simulation.

Numerical Solving

The governing equations of the electrochemical regeneration model were implemented in Python and solved as a transient initial-value problem using `solve_ivp` with the RK45 integration method. The system is formulated in terms of nine time-dependent state variables, namely the moles of dissolved inorganic carbon, sodium in the acid and base tanks, hydroxide in the base tank, released CO_2 , residual NaOH in the acid tank, water in both tanks, and the lumped system temperature.

At each integration step, the model recalculates tank volumes from the instantaneous species inventories and then updates the corresponding concentrations, carbonate speciation, activity-corrected pH values, membrane transport fluxes, conductivity, electrode overpotentials, and thermal terms. In this way, the transport, electrochemical, and thermodynamic submodels remain fully coupled throughout the transient simulation.

The time integration is terminated either when the prescribed operating time is reached or when the acid-tank pH falls to the specified stopping criterion, which is implemented in the code as a terminal event function. After integration, the solution is post-processed to obtain the temporal profiles of species concentrations, pH in both compartments, NaOH production, CO_2 release, cell voltage, power demand, conductivity, transport numbers, and material-balance and energy-performance indicators.

Results and Discussion

Suggestion: Validation

The absorber model showed excellent internal consistency, with CO_2 and H_2O balance errors of only 4.18×10^{-6} and 1.21×10^{-9} , respectively, while the limiting-case analysis gave 3.78% capture at $Z=0.1$ m and 100% at $Z=30$ m, confirming physically correct behavior. The axial profiles were also monotonic, the BVP converged exactly, and the NTU-HTU relation was satisfied (NTU=5.685, HTU=2.638 m, NTU×HTU=15.0 m) (NTU=5.685, HTU=2.638 m, NTU×HTU=15.0 m), which matches the column height.

External validation[41,42] shows that the absorber model is consistent with published NaOH-based DAC behavior. Ghaffari et al. studied 0.0153–1.2 vol% CO_2 at 5–20 °C and reported overall mass-transfer coefficients of about 0.5–3 mm s⁻¹, with Hatta numbers always above 3, supporting the fast-reaction regime predicted by the model. Seithümmel et al. also reported strongly alkaline counter-current NaOH absorption, with carbonate-favored chemistry above about pH 13.5 and pH inflection points around

11.5–12.0, which is consistent with the model's high-pH, carbonate-dominated operation.

Absorber Column characteristics

The isothermal absorber exhibits strong CO₂ removal performance under the selected base-case conditions. The auto-sized column has a diameter of 29 cm and a packed height of 15 m, with gas and liquid superficial velocities of 14.7 cm s⁻¹ and 10.09 mm s⁻¹, respectively. Hydraulically, the column operates in a safe regime at about 65.1% of the flooding velocity and shows only a small pressure drop of 51.2 Pa, corresponding to 0.03 mbar m⁻¹. The overall mass-transfer performance is high, with $k_G a_w = 2.585 \text{ mol m}^{-3} \text{ s}^{-1} \text{ atm}^{-1}$, and the transfer is mainly gas-film controlled because about 90.2% of the total resistance lies in the gas phase. As a result, the inlet CO₂ concentration of 420 ppm is reduced to about 1.43 ppm at the outlet, corresponding to a capture efficiency of 99.66%. At the same time, the solvent composition changes only slightly along the column, since the free NaOH concentration remains close to 2 mol L⁻¹, the pH stays around 14.23, and the NaOH utilization is only 0.03%, indicating operation with a large excess of caustic solution. The model also predicts noticeable water evaporation, with the gas-phase water content increasing from 1.379% at the inlet to 2.217% at the outlet and an evaporative duty of 152.7 W, which is larger than the reaction heat release of 18.4 W.

The figure 1 & 2 explain this behavior in more detail. In the top graph, both the gas-phase CO₂ mole fraction y_{CO_2} and the dissolved inorganic carbon concentration C_{DIC} decrease along the normalized column coordinate ζ , which runs from $\zeta=0$ at the gas inlet and liquid outlet to $\zeta=1$ at the gas outlet and liquid inlet. The steep decline near the lower section of the column shows that most of the CO₂ absorption takes place close to the gas inlet, while both profiles approach very small values toward the top of the absorber. The bottom graph shows that the local absorption flux N_{CO_2} and the driving force ($y_{\text{CO}_2} - y_{\text{CO}_2}^*$) follow the same axial trend, with their highest values near $\zeta=0$ and a continuous decrease toward $\zeta=1$. This indicates that mass transfer is strongest where the entering gas still contains the highest CO₂ concentration and therefore provides the largest driving force for absorption. The close similarity of the two lower curves further shows that the decline in absorption rate is governed primarily by the progressive reduction in the available driving force along the column.

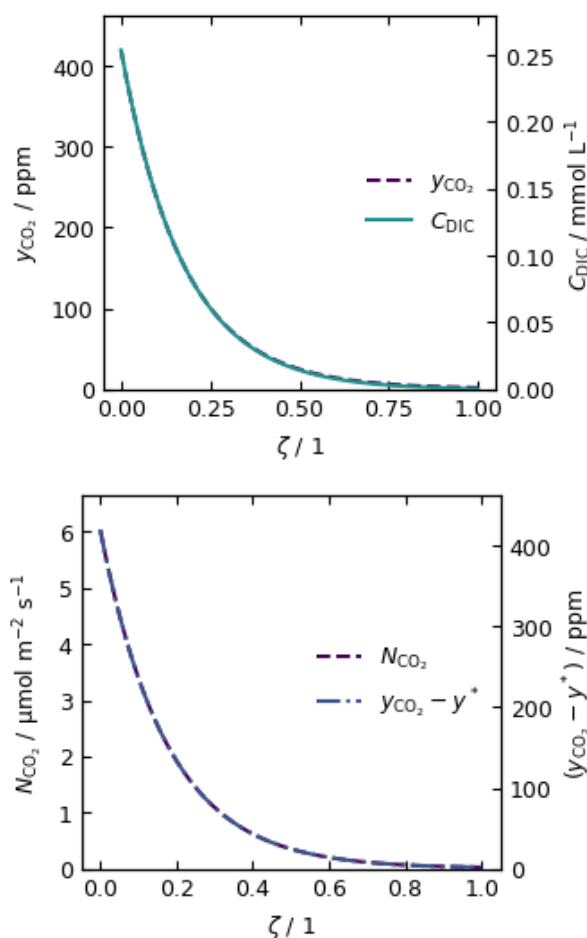


Figure 1 & 2: Top – Axial profiles of gas phase CO₂ mole fraction, y_{CO_2} and dissolved inorganic carbon concentration, C_{DIC} along the normalized absorber height, ζ . Bottom – Axial profiles of local CO₂ molar flux, N_{CO_2} , and gas-phase driving force, $(y_{\text{CO}_2} - y_{\text{CO}_2}^*)$, along the normalized absorber height, ζ .

Sensitivity Analysis of Absorber column

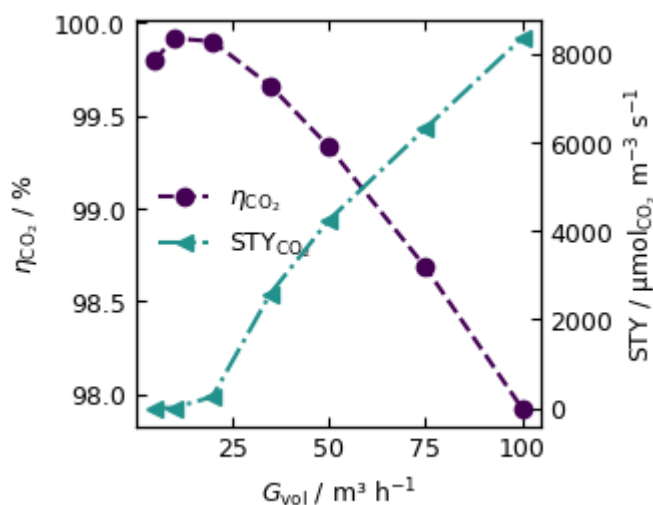


Figure 3: Effect of gas volumetric flow rate, G_{vol} , on CO₂ capture efficiency, η_{CO_2} , and space-time yield (STY) of the absorber column.

The first parameter varied in the sensitivity analysis is the gas volumetric flow rate G_{vol} , investigated between 5 and 100 $\text{m}^3 \text{h}^{-1}$. Figure 3 shows that the CO_2 capture efficiency and the space-time yield respond in opposite ways to increasing gas throughput. At low gas flow rates, the absorber achieves nearly complete CO_2 removal because the gas residence time is long, but the STY remains low due to the limited amount of processed gas. As G_{vol} increases, the STY rises strongly because more CO_2 -containing gas is handled per unit absorber volume and time, whereas the capture efficiency declines gradually as the available contact time decreases. The standard case at 35 $\text{m}^3 \text{h}^{-1}$ provides a useful reference point, yielding 99.66% capture efficiency and an STY of 2585 $\mu\text{mol m}^{-3} \text{s}^{-1}$. In addition, the detailed hydraulic results show that the lowest-flow cases are associated with excessively large auto-sized diameters and very low $k_G a_w$, while higher-flow cases shift toward a liquid-loading-controlled design with substantially higher gas-side mass-transfer capacity. Thus, the gas flow-rate sensitivity highlights the central design trade-off of the absorber: low gas flow favors removal efficiency, whereas high gas flow favors productivity.

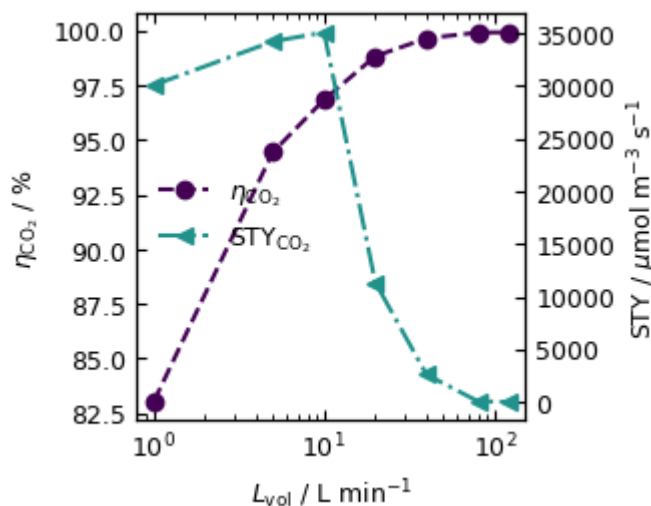


Figure 4: Effect of liquid volumetric flow rate, L_{vol} , on CO_2 capture efficiency, η_{CO_2} , and space-time yield (STY) of the absorber column.

The second parameter varied in the sensitivity analysis was the liquid volumetric flow rate, L_{vol} , over the range 1–120 L min^{-1} . Figure 4 shows that increasing L_{vol} substantially improves the CO_2 capture efficiency, which rises from 83.02% at 1 L min^{-1} to 99.93% at 120 L min^{-1} . This indicates that higher solvent throughput enhances CO_2 absorption by maintaining a stronger driving force and greater liquid-side capacity along the column.

In contrast, the STY shows a non-monotonic trend with liquid flow rate. It increases from 30,085 $\mu\text{mol m}^{-3} \text{s}^{-1}$ at 1 L min^{-1} to a maximum of 35,097 $\mu\text{mol m}^{-3} \text{s}^{-1}$ at 10 L min^{-1} , and then decreases sharply to 11,333, 2585, 57.9, and 1.5 $\mu\text{mol m}^{-3} \text{s}^{-1}$ at 20, 40, 80, and 120 L min^{-1} , respectively. Although larger liquid flow rates improve capture efficiency, they strongly reduce absorber productivity beyond the intermediate range.

This behavior is explained by the hydraulic resizing of the absorber. Up to 10 L min^{-1} , the column diameter remains 15 cm, the superficial gas velocity stays high at about 55.0 cm s^{-1} , and $k_G a_w$ increases to 6.834 $\text{mol m}^{-3} \text{s}^{-1} \text{atm}^{-1}$, which supports high STY. At higher liquid flow

rates, the absorber becomes liquid-loading or gas-flooding controlled, the diameter increases from 20 cm at 20 L min^{-1} to 187 cm at 120 L min^{-1} , and $k_G a_w$ drops to 0.072 $\text{mol m}^{-3} \text{s}^{-1} \text{atm}^{-1}$, leading to much weaker gas-side mass transfer and very low productivity.

Overall, the results show that higher liquid flow rates favor near-complete CO_2 removal, whereas intermediate flow rates around 5–10 L min^{-1} provide the best compromise between capture efficiency and STY. Very high liquid flow rates mainly improve the final fraction of CO_2 capture, but only at the expense of severe productivity loss.

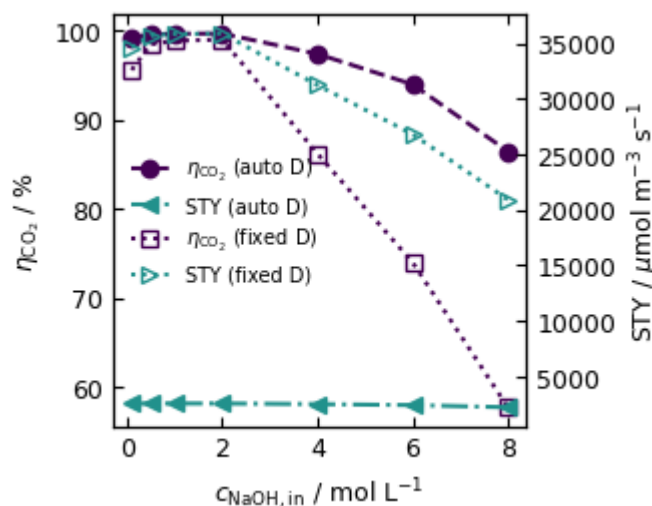


Figure 5: Effect of inlet NaOH concentration, $C_{\text{NaOH},in}$, on CO_2 capture efficiency, η_{CO_2} , and space-time yield (STY) for both the auto-sized and fixed-diameter absorber configurations.

The third parameter varied in the sensitivity analysis was the inlet NaOH concentration, $C_{\text{NaOH},in}$, over the range 0.1–8.0 mol L^{-1} . Figure 5 shows that, under the auto-sized design, increasing the NaOH concentration from 0.1 to 2.0 mol L^{-1} produces only a modest improvement in performance, with the capture efficiency increasing from 99.11% to 99.66% and the STY remaining nearly constant at about 2571–2585 $\mu\text{mol m}^{-3} \text{s}^{-1}$. This indicates that, within the low-to-moderate concentration range, the absorber is already operating close to its optimum and further increases in NaOH concentration provide only marginal benefit.

At higher NaOH concentrations, however, the absorber performance deteriorates noticeably. In the auto-sized cases, increasing $C_{\text{NaOH},in}$ from 2.0 to 8.0 mol L^{-1} reduces the capture efficiency from 99.66% to 86.45% and lowers the STY from 2585 to 2242 $\mu\text{mol m}^{-3} \text{s}^{-1}$. The same qualitative trend is observed in the fixed-diameter runs, where the capture efficiency drops from 98.99% at 2.0 mol L^{-1} to 57.73% at 8.0 mol L^{-1} , while the STY decreases from 35,873 to 20,921 $\mu\text{mol m}^{-3} \text{s}^{-1}$. Since the decline appears in both the auto-sized and fixed-diameter results, the negative effect of high NaOH concentration is not merely an artifact of column resizing.

Optimization of Absorber Column

The absorber operating window was optimized by varying the inlet NaOH concentration, $C_{\text{NaOH},in}$ and the liquid volumetric flow rate, L_{vol} , and evaluating the resulting space-time yield (STY) and CO_2 capture efficiency. The contour plots (figure 6 and 7) show that the two objectives do not coincide exactly, so the preferred operating point

depends on whether the design target is maximum productivity or maximum capture efficiency.

The STY map indicates that productivity is maximized in the low-to-moderate liquid-flow region, with the global maximum located at $c_{\text{NaOH},\text{in}}=2.00 \text{ mol L}^{-1}$ and $L_{\text{vol}}=10 \text{ L min}^{-1}$, where the STY reaches $35,096.8 \mu\text{mol m}^{-3} \text{ s}^{-1}$. More generally, the high-STY region forms a broad plateau at moderate NaOH concentrations of roughly $0.5\text{--}3 \text{ mol L}^{-1}$ and low liquid flow rates around $5\text{--}15 \text{ L min}^{-1}$. Outside this region, STY decreases sharply with increasing liquid flow rate, falling to very low values at $60\text{--}80 \text{ L min}^{-1}$, which shows that absorber productivity is strongly penalized by excessive solvent circulation. see figure 6.

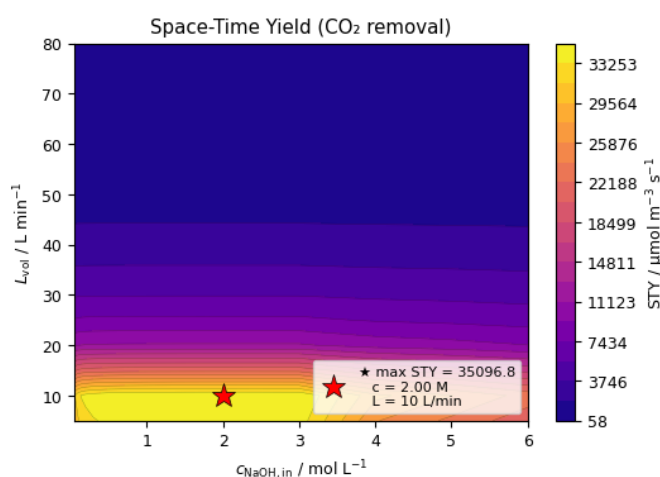


Figure 6: Contour plot of space-time yield (STY) as a function of inlet NaOH concentration, $c_{\text{NaOH},\text{in}}$, and liquid volumetric flow rate, L_{vol} .

The capture-efficiency map shows a different optimum. The maximum capture efficiency is 99.92%, obtained at $c_{\text{NaOH},\text{in}}=2.00 \text{ mol L}^{-1}$ and $L_{\text{vol}}=80 \text{ L min}^{-1}$. The contour field further shown in figure 7 indicates that very high CO_2 removal can be maintained over a broad operating window at moderate NaOH concentrations, whereas efficiency deteriorates at high NaOH concentrations, particularly above about 4 mol L^{-1} , and most strongly at low liquid flow rates.

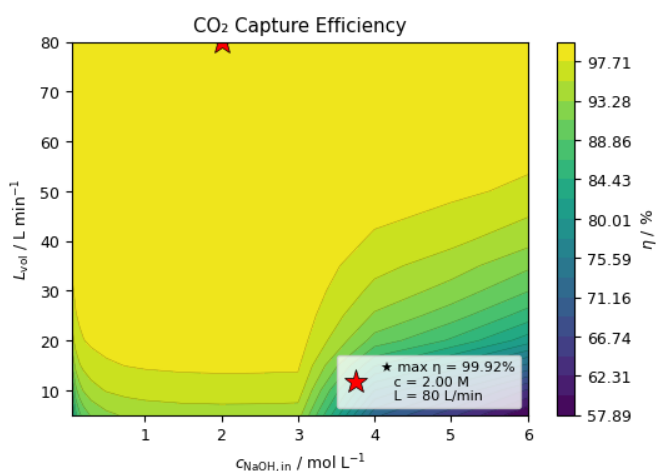


Figure 7: Contour plot of CO_2 capture efficiency, η , as a function of inlet NaOH concentration, $c_{\text{NaOH},\text{in}}$, and liquid volumetric flow rate, L_{vol} .

Taken together, the two plots suggest that an NaOH concentration near 2 mol L^{-1} is a robust choice, since it appears at the optimum of both objective functions. The main trade-off is therefore governed by the liquid flow rate: low L_{vol} favors maximum STY, while high L_{vol} favors near-complete CO_2 capture. For practical absorber design, the region around $c_{\text{NaOH},\text{in}} \approx 2 \text{ mol L}^{-1}$ and $L_{\text{vol}} = 10\text{--}20 \text{ L min}^{-1}$ appears to offer the most balanced compromise, because it preserves very high capture efficiency while retaining much higher productivity than the high-flow operating range.

Electrochemical Cell Characteristics

The electrochemical cell model represents a seven-cell Nafion 117 stack (total area 0.35 m^2) that electrochemically regenerates NaOH from a carbonate-rich DAC absorber effluent while producing a concentrated CO_2 stream. At the matched operating point, the absorber converts an initial 2.0 M NaOH solution (200 L) to 0.950 M $\text{Na}_2\text{CO}_3 + 0.100 \text{ M}$ NaOH over 331 h , corresponding to 190 mol (8.36 kg) CO_2 captured in the liquid phase. The EC cell is then operated at a current density of 92.4 A m^{-2} (9.2 mA cm^{-2}) with a per-cell current of 4.62 A and an average cell voltage of 0.711 V , giving a stack power of 23 W and total electrical power consumption (including pumps and PSU losses) of 26.7 W . Over a 331-h regeneration cycle, the model predicts 183.3 mol CO_2 desorbed (96.5% recovery), production of 214.5 L of 1.72 M NaOH product at 92% purity, and an H_2 byproduct stream of 183.7 mol that is fully recycled. The specific energy consumption is $160 \text{ kJ mol}^{-1} \text{ CO}_2$ (1012 kWh t^{-1}), which lies near the theoretical minimum of 164 kJ mol^{-1} reported by Shu et al., indicating that the model is thermodynamically consistent while still realistically accounting for kinetic and ohmic losses. The Nernst–Planck membrane sub-model yields Na^+ transport numbers decreasing from 1.000 to 0.9982 and H^+ transport numbers increasing from 0 to 0.0018 , with a membrane potential of 40.7 mV and conductivity rising from 0.5 to 0.6 S m^{-1} , and the overall voltage breakdown shows moderate activation overpotentials and ohmic drops that are compatible with literature data for Nafion-based alkaline regeneration cells.

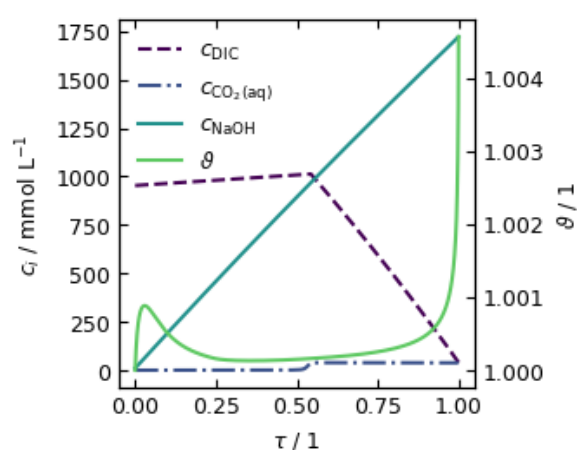


Figure 8: composition and volume vs. normalized time

The figure 8 shows the normalized cycle time $\tau/1$ on the x-axis ($0 = \text{start}$, $1 = \text{pH-stop}$) and three main quantities on the left y-axis: total dissolved inorganic carbon (DIC), dissolved $\text{CO}_2(\text{aq})$, and NaOH concentration. DIC increases almost linearly to about 1700 mmol L^{-1}

as Na_2CO_3 is progressively converted and carbon species are driven to the acid side, while NaOH in the base compartment rises from its seed level to about 1.72 M as regeneration proceeds. The $\text{CO}_2(\text{aq})$ curve stays low and only spikes near the end of the cycle, reflecting that most CO_2 leaves as gas rather than remaining dissolved until the pH-stop is approached. The right y-axis shows the relative volume change $\vartheta = V/V_0$, which remains close to unity for most of the cycle and then increases sharply near $\tau \approx 1$, indicating the late-cycle volume expansion of the base tank due to water drag and NaOH accumulation.

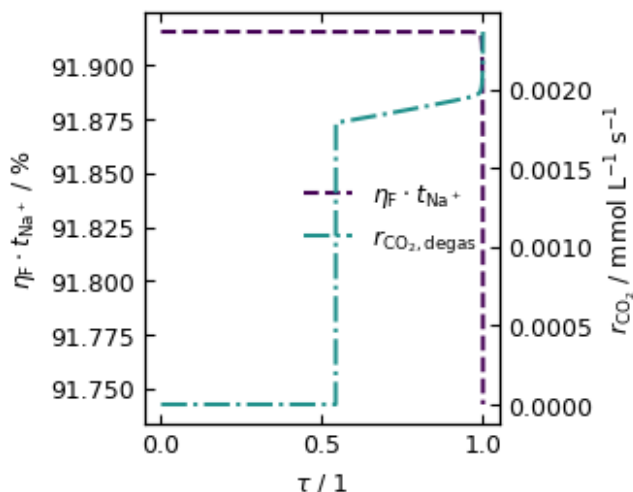


Figure 9: current efficiency and CO_2 evolution rate

The figure 9 again uses normalized time on the x-axis and shows the total current efficiency for Na^+ transfer, η_t , on the left y-axis and the instantaneous CO_2 evolution rate $r_{\text{CO}_2, \text{degas}}$ on the right y-axis. The faradaic efficiency remains very high and nearly constant (around 91.8–91.9%) over most of the cycle, confirming that almost all charge goes into the desired NaOH-generation pathway. Near the end of the cycle, both curves exhibit a sharp change: η_t shows a step as the system approaches the pH-stop and carbonate is nearly depleted, while $r_{\text{CO}_2, \text{degas}}$ jumps up and then drops to zero at $\tau = 1$, capturing the final burst of CO_2 release before the cell is switched off.

Capex and Opex

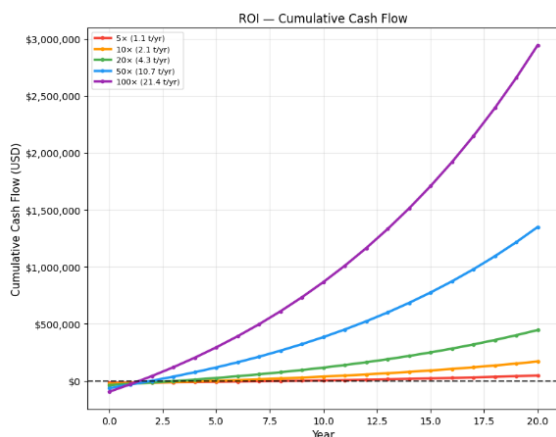


Figure 10: Cumulative cash flow over the years by scaling up

At current lab scale (1 $\times$), the plant is not economically viable. However, scaling to just 5 $\times$ (1.1 ton CO_2 /yr, CAPEX $\sim$ \$16,200) achieves profitability with 8.6-year payback when selling food-grade CO_2 in Pakistan's premium domestic market. At 50 $\times$ (10.7 ton/yr), the payback drops to 2.0 years with NPV of \$410,000 over 20 years, making it a commercially attractive investment. A detail table on return on investment is given in the appendix.

Conclusions

This work developed and evaluated a coupled DAC process using NaOH absorption and electrochemical regeneration. The absorber model showed that very high CO_2 removal can be achieved under ambient-air conditions, with the base case giving 99.66% capture and reducing the outlet concentration to only 1.43 ppm. The hydraulic results also indicated stable operation, with moderate flooding, low pressure drop, and physically consistent axial concentration profiles, while the sensitivity analysis showed clear trade-offs between capture efficiency and productivity. In particular, higher liquid flow rates improved capture efficiency but reduced space-time yield, whereas an NaOH concentration around 2 mol L^{-1} gave the best overall performance.

The electrochemical regeneration model showed that the carbonate-rich solution from the absorber can be converted back into a useful NaOH stream while releasing a concentrated CO_2 product. For the integrated base case, the model predicted 96.5% CO_2 recovery, a regenerated NaOH concentration of 1.72 mol L^{-1} , and a product purity of 92.0%. The calculated specific energy demand of 160 kJ mol $^{-1}$ CO_2 is close to the theoretical minimum reported in the literature, which suggests that the process is energetically promising. The cycle simulation also showed a good match between absorber loading time and regeneration time, indicating that the two units can be operated together in a practical way.

Overall, the results suggest that electrochemical regeneration could be a viable alternative to conventional thermal regeneration in alkaline direct air capture. The developed model provides a useful basis for further optimization of solvent utilization, energy demand, and equipment sizing. Future work should focus on developing a dynamic absorber model for cyclic operation, since the present study used a surrogate absorber approach to estimate the number of cycles needed to reach the feed composition of the electrochemical cell. Such a model should include the same design factors as the detailed single-cycle absorber, allowing more accurate prediction of cycle number and other transient effects in the integrated process. In addition, because regeneration is faster than absorption, the use of multiple absorber columns could be explored to improve process matching, although this would involve a trade-off between higher capital cost and better operational performance.

Supporting Information

Both models for Absorber column and electrochemical cell are given as zip folder. Sensitivity analysis for electrochemical cell was also done but not included because of word count limit but it is in the jupyter notebook file DAC_Ecell. Detailed Capex and Opex study are also provided as ipynb file.

Author Contributions

The entire work was done by Muhammad Arslan Aslam only. The other two members of the group did not contribute to the project in any capacity through out the semester.

Statutory Declaration

I declare that I have authored this report independently, that I have not used other than the declared sources, and that I have explicitly marked all material which has been quoted either literally or by content from the used sources.

Acknowledgements

I would like to thank M.Sc. Tuana Oyuncu and M.Sc. Theresa Kunz for the ideas and feedback during consultations.

Nomenclature

Symbol	Unit	Description
A_{cell}	m ²	Active area of one electrochemical cell
A_{tot}	m ²	Total active area, $A_{tot} = A_{cell} N_{cell}$
C_{DIC}	mol L ⁻¹	Dissolved inorganic carbon concentration in absorber model
$C_{p,G}$	J mol ⁻¹ K ⁻¹	Gas-phase molar heat capacity
$C_{p,L}$	J L ⁻¹ K ⁻¹	Liquid-phase volumetric heat capacity
$C_{p,a}$	J kg ⁻¹ K ⁻¹	Heat capacity of acid-side liquid
$C_{p,b}$	J kg ⁻¹ K ⁻¹	Heat capacity of base-side liquid
D_A	m ² s ⁻¹	Diffusivity of dissolved CO ₂
E	–	Enhancement factor
E_{cell}	V	Total cell voltage
E_{rev}	V	Reversible voltage
$E_{rev,act}$	V	Activity-corrected reversible voltage
F	C mol ⁻¹	Faraday constant
G_{inert}	mol m ⁻² s ⁻¹	Inert-gas molar flux
G_{mol}	mol m ⁻² s ⁻¹	Local gas molar flux
G_{vol}	m ³ h ⁻¹	Gas volumetric flow rate
H_{eff}	mol L ⁻¹ atm ⁻¹	Effective henry coefficient
H_a	–	Hatta number
I_{tot}	A	Total current
K_G	mol m ⁻² s ⁻¹ atm ⁻¹	Overall gas-side mass-transfer coefficient
L_{vol}	L min ⁻¹	Liquid volumetric flow rate
N_{cell}	–	Number of cells
N_{CO2}	mol m ⁻² s ⁻¹	CO ₂ molar flux
N_{H2O}	mol m ⁻² s ⁻¹	Water vapor flux
P	W	Electrical power
Q_{GL}	J mol ⁻¹	Interphase heat transfer
Q_{evap}	J mol ⁻¹	Latent heat evaporation
Q_{rxn}	J mol ⁻¹	Heat of reaction
R	J mol ⁻¹ K ⁻¹	Gas constant
$R_{H,back}$	mol s ⁻¹	Proton back-transport
$R_{H2O,drag}$	mol s ⁻¹	Water drag

R_{Na}	mol s ⁻¹	Sodium transport
R_{degas}	mol s ⁻¹	CO ₂ degassing
$R_{neutralize}$	mol s ⁻¹	Neutralization rate
SEC	kJ mol ⁻¹ CO ₂	Specific energy consumption
STY	μmol m ⁻³ s ⁻¹	Space-time yield
T	K	System temperature
T_G	K	Gas temperature
T_L	K	Liquid temperature
T_{amb}	K	Ambient temperature
UA_{ext}	W K ⁻¹	Heat transfer parameter
V_a	L	Acid tank volume
V_b	L	Base tank volume
Y_{CO2}	–	CO ₂ mole ratio
Y_{H2O}	–	Water mole ratio
C_{DIC}	mol L ⁻¹	DIC concentration (surrogate)
C_{NaOH}	mol L ⁻¹	Inlet naoh concentration
$C_{i,a}$	mol L ⁻¹	Species conc. In acid tank
$C_{i,b}$	mol L ⁻¹	Species conc. In base tank
C_{OH}	mol L ⁻¹	Hydroxide concentration
f_{ion}	–	Ionic strength factor
$J_{applied}$	A m ⁻²	Applied current density
$J_{parasitic}$	A m ⁻²	Parasitic current density
k_G	mol m ⁻² s ⁻¹ atm ⁻¹	Gas-side coefficient
k_L	m s ⁻¹	Liquid-side coefficient
k_{OH}	L mol ⁻¹ s ⁻¹	Reaction rate constant
k_{app}	s ⁻¹	Apparent rate constant
$k_G a_w$	mol m ⁻³ s ⁻¹ atm ⁻¹	Volumetric mass-transfer
m_a	kg	mass acid
m_b	kg	mass base
$n_{CO2,g}$	mol	Released CO ₂
n_{DIC}	mol	DIC inventory
$n_{H2O,a}$	mol	Water acid tank
$n_{H2O,b}$	mol	Water base tank
$n_{Na,a}$	mol	Na acid tank
$n_{Na,b}$	mol	Na base tank
$n_{OH,b}$	mol	OH base tank
n_{res}	mol	Residual NaOH
pH_a	–	pH acid
pH_b	–	pH base
$r_{co2,dega}$	mol L ⁻¹ s ⁻¹	CO ₂ evolution rate
r_e	mol s ⁻¹	Electron transfer rate
u_L	M S ⁻¹	Liquid velocity
$u_{L,loc}$	m s ⁻¹	Local velocity
y	–	State vector
y_{co2}	–	CO ₂ mole fraction
y_{co2}^*	–	Equilibrium CO ₂ fraction
z	m	Axial coordinate
Z	m	Packed height

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Appendix

The following equations summarize the additional constitutive, thermodynamic, hydraulic, and transport relations used in the absorber model.

Composition relations [11]

The gas-phase balances are solved in terms of mole ratios, whereas thermodynamic and transfer calculations are carried out using gas-phase mole fractions. The conversion relations are given by Eqs. (A1) -(A4).

$$y_{CO_2} = \frac{Y_{CO_2}}{(1 + Y_{CO_2} + Y_{H_2O})} \quad (A1)$$

$$y_{H_2O} = \frac{Y_{H_2O}}{(1 + Y_{CO_2} + Y_{H_2O})} \quad (A2)$$

$$Y_{CO_2} = \frac{y_{CO_2}}{(1 - y_{CO_2} - y_{H_2O})} \quad (A3)$$

$$Y_{H_2O} = \frac{y_{H_2O}}{(1 - y_{CO_2} - y_{H_2O})} \quad (A4)$$

Local liquid-velocity correction

The local liquid superficial velocity used in the DIC balance is corrected as a function of dissolved inorganic carbon loading.

$$u_{L,loc} = u_L [1 + 0.033 \max(C_{DIC} - C_{DIC,top}, 0)] \quad (A5)$$

Carbonate speciation [12,13]

The liquid-phase carbonate system is described by dissolved inorganic carbon conservation, acid-base equilibria, and electroneutrality. The corresponding equations are given by Eqs. (A6) -(A15).

$$C_{DIC} = c_{CO_2,aq} + c_{HCO_3^-} + c_{CO_3^{2-}} \quad (A6)$$

$$K_1 = 10^{-(6.352 - 0.00317(T - 298.15))} \quad (A7)$$

$$K_2 = 10^{-(10.329 - 0.00418(T - 298.15))} \quad (A8)$$

$$K_w = 10^{-(13.833 + 0.0183(298.15 - T))} \quad (A9)$$

$$c_{HCO_3^-} = \left(\frac{K_1}{h}\right) c_{CO_2,aq} \quad (A10)$$

$$c_{CO_3^{2-}} = \left(\frac{K_2}{h}\right) c_{HCO_3^-} \quad (A11)$$

$$c_{OH^-} = \frac{K_w}{h} \quad (A12)$$

$$(c_{Na,tot} + h) - (c_{OH^-} + c_{HCO_3^-} + 2 c_{CO_3^{2-}}) = 0 \quad (A13)$$

$$pH = -\log_{10}(h) \quad (A14)$$

$$c_{NaOH,free} = c_{Na,tot} - c_{HCO_3^-} - 2 c_{CO_3^{2-}} \quad (A15)$$

CO2 solubility and equilibrium [14,15,16,44]

The equilibrium relation for CO2 absorption is evaluated through a Henry coefficient corrected for ionic strength in alkaline solution. The relevant expressions are given by Eqs. (A16) -(A19).

$$I = \max(c_{NaOH}, 0) + 3 \max(c_{Na_2CO_3}, 0) \quad (A16)$$

$$H_{eff} = \frac{H_{CO_2,w}}{10^{0.11I}} \quad (A17)$$

$$y^*_{CO_2} = \frac{\left(\frac{c_{CO_2,aq}}{H_{eff}}\right)}{P/101325} \quad (A18)$$

Infinite enhancement limit [17]

The infinite-enhancement limit used in the enhancement-factor model is calculated from Eqs. (A20) and (A21).

$$c_{A,i} = H_{eff} y_{CO_2} \left(\frac{P}{101325}\right) \quad (A20)$$

$$E_\infty = 1 + \left(\frac{D_B}{D_A}\right) \left(\frac{c_{OH^-}}{2 c_{A,i}}\right) \quad (A21)$$

Gas- and liquid-side mass-transfer coefficients [10]

The gas-side mass-transfer coefficient is evaluated using a Billet-Schultes-type correlation, as shown in Eq. (A22).

$$k_{G,BS} = C_{BG} \sqrt{\left(\frac{a_p}{\varepsilon - h_L}\right)} D_G \left(\frac{Re_G}{\varepsilon}\right)^{0.75} Sc_G^{\frac{1}{3}} \left(\frac{P}{R T_G}\right) \quad (A22)$$

The liquid-side mass-transfer coefficient can be calculated with Billet-Schultes, Higbie, or Onda formulations, as given by Eqs. (A23) -(A25). [10],[43]

$$k_{L,BS} = C_{BL} \sqrt{\left(\frac{D_L}{d_h}\right)} \sqrt{(u_L, eff)} \quad (A23)$$

$$k_{L,Higbie} = 2 \sqrt{\left(\frac{D_L}{\pi \tau_{exp}}\right)} \quad (A24)$$

$$k_{L,Onda} = 0.0051 Re_L^{\frac{2}{3}} Sc_L^{-\frac{1}{2}} (a_p d_p)^{0.4} \left(\frac{\mu_L g}{\rho_L}\right)^{\frac{1}{3}} \quad (A25)$$

Hydraulic and wetting relations [10],[40]

The hydraulic submodel includes liquid holdup, effective voidage, wetted area, flooding velocity, and pressure drop. These relations are summarized in Eqs. (A26) -(A32).

$$h_L = \left[\frac{12 \mu_L u_L a_p^2}{(\rho_L g)} \right]^{\frac{1}{3}} \quad (A26)$$

$$\varepsilon_{eff} = \varepsilon - h_L \quad (A27)$$

$$Re_L = \frac{\rho_L u_L}{\mu_L a_p}; Fr_L = \frac{u_L^2 a_p}{g}; We_L = \frac{\rho_L u_L^2}{\sigma_L a_p} \quad (A28)$$

$$a_w = a_p \left[1 - \exp \left(-1.45 \left(\frac{\sigma_c}{\sigma_L} \right)^{0.75} Re_L^{0.1} Fr_L^{-0.05} We_L^{0.2} \right) \right] \quad (A29)$$

$$u_{G,flood} = C_v \left[\frac{g \varepsilon_{eff}^3}{a_p \left(\frac{\rho_G}{\rho_L} \right)} \right]^{\frac{1}{2}} \exp(-1.463 F_{lv}^{0.842}) \quad (A30)$$

$$\left(\frac{dP}{dz} \right)_{dry} = \frac{C_h \left(\frac{a_p}{\varepsilon^3} \right) \rho_G u_G^2}{2} \quad (A31)$$

$$\left(\frac{dP}{dz} \right)_{wet} = \left(\frac{dP}{dz} \right)_{dry} \left[1 + 0.2 \left(\frac{u_L}{u_G} \right)^{0.3} \right] \quad (A32)$$

Water evaporation [18,19]

Water evaporation from the alkaline solvent is represented by a vapour-pressure-driven gas-side transfer relation. The governing auxiliary expressions are given by Eqs. (A33) -(A37).

$$m = \frac{c_{NaOH}}{\left(\frac{\rho_L}{1000} - 0.040 c_{NaOH}\right)} \quad (A33)$$

$$a_w^{water} = \max(1 - 0.017 m, 0.5) \quad (A34)$$

$$p_{sat,NaOH} = a_w^{water} p_{sat,water} \quad (A35)$$

$$k_{G,H_2O} = k_G \left(\frac{D_{H_2O,air}}{D_{CO_2,air}} \right)^{\frac{2}{3}} \quad (A36)$$

$$N_{H_2O} = k_{G,H_2O} \left(\frac{P}{101325} \right) \left(\frac{p_{sat,NaOH}}{P} - y_{H_2O} \right) \quad (A37)$$

Reaction enthalpy model [21,22]

The pH-dependent heat of reaction used in the adiabatic model is defined by Eqs. (A38) and (A39). [21,22]

$$\Delta H_{rxn} = f_1(-109400) + (1 - f_1)(-15100) \quad (A38)$$

$$f_1 = \frac{1}{1 + \exp\left(\frac{11.74 - pH}{0.5}\right)} \quad (A39)$$

Auxiliary transport and property equations [20,21]

The gas density is calculated from the ideal-gas relation, as shown in Eq. (A40).

$$\rho_G = \frac{PM_{air}}{RT} \quad (A40)$$

The binary gas diffusivities used in the model are given by Eqs. (A41) and (A42).

$$D_{CO_2,air} = 1 \times 10^{-7} \frac{T^{1.75} \sqrt{\left(\frac{1}{44} + \frac{1}{29}\right)}}{\left(\frac{P}{101325}\right) \left(26.9^{\frac{1}{3}} + 19.7^{\frac{1}{3}}\right)^2} \quad (A41)$$

$$D_{H_2O,air} = 1 \times 10^{-7} \frac{T^{1.75} \sqrt{\left(\frac{1}{18} + \frac{1}{29}\right)}}{\left(\frac{P}{101325}\right) \left(13.1^{\frac{1}{3}} + 19.7^{\frac{1}{3}}\right)^2} \quad (A42)$$

The liquid-phase diffusivities are calculated from Eqs. (A43) and (A44).

$$D_{CO_2,liq} = 1.92 \times 10^{-9} \left(\frac{T}{298.15} \right) \exp \left[-1600 \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] \exp(-0.12 c_{NaOH}) \quad (A43)$$

$$D_{OH^-,liq} = 5.27 \times 10^{-9} \left(\frac{T}{298.15} \right) \quad (A44)$$

The gas and liquid heat capacities are represented by simplified composition-dependent relations, given in Eqs. (A45) and (A46). [19]

$$C_{p,G} = 37.1 y_{CO_2} + 29.1 y_{N_2} + 29.4 y_{O_2} + 20.8 y_{Ar} + 33.6 y_{H_2O} \quad (A45)$$

$$C_{p,L} = 4186 - 20 (c_{NaOH} + c_{Na_2CO_3}) \quad (A46)$$

Appendix B

Constitutive Equations and Auxiliary Relations used in electrochemical cell model.

Volume and concentration relations

The acid and base tank volumes are calculated from the instantaneous species inventories and partial molar volumes. In the implementation, lower bounds are imposed on the calculated volumes for numerical robustness during time integration.

$$V_a = \max(n_{H_2O,a} \bar{V}_{H_2O} + n_{Na,a} \bar{V}_{Na^+} + n_{DIC} \bar{V}_{CO_2}, 10) \quad (B1)$$

$$V_b = \max(n_{H_2O,b} \bar{V}_{H_2O} + n_{Na,b} \bar{V}_{Na^+} + n_{OH,b} \bar{V}_{OH^-}, 10) \quad (B2)$$

$$c_{DIC} = \frac{n_{DIC}}{V_a}, c_{Na,a} = \frac{n_{Na,a}}{V_a}, c_{Na,b} = \frac{n_{Na,b}}{V_b}, c_{OH,b} = \frac{n_{OH,b}}{V_b}, c_{res} = \frac{n_{res}}{V_a} \quad (B3)$$

Carbonate speciation in the acid tank

The acid-tank carbonate system is described by dissolved inorganic carbon conservation, water dissociation, carbonate equilibria, and electroneutrality. The temperature-dependent equilibrium constants used in the code are [45]

$$K_1(T) = 10^{-\left(\frac{3404.71}{T} + 0.03279 T - 14.8435\right)} \quad (B4)$$

$$K_2(T) = 10^{-\left(\frac{2902.39}{T} + 0.02379 T - 6.4980\right)} \quad (B5)$$

$$K_w(T) = 10^{-\left(\frac{4470.99}{T} + 0.01706 T - 6.0846\right)} \quad (B6)$$

Using $h = [H^+]$, the speciation relations are

$$d = 1 + \frac{K_1}{h} + \frac{K_1 K_2}{h^2} \quad (B7)$$

$$c_{CO_2,aq} = \frac{C_{DIC}}{d} \quad (B8)$$

$$c_{HCO_3^-} = c_{CO_2,aq} \frac{K_1}{h} \quad (B9)$$

$$c_{CO_3^{2-}} = c_{HCO_3^-} \frac{K_2}{h} \quad (B10)$$

$$c_{OH^-} = \frac{K_w}{h} \quad (B11)$$

$$(c_{Na,a} + h) - (c_{OH^-} + c_{res} + c_{HCO_3^-} + 2c_{CO_3^{2-}}) = 0 \quad (B12)$$

$$pH_a = -\log_{10}(h) \quad (B13)$$

The base-tank relations are

$$h_b = \frac{K_w}{c_{OH,b}} \quad (B14)$$

$$pH_b = -\log_{10}(h_b) \quad (B15)$$

$$c_{NaOH,prod} = \min(c_{Na,b}, c_{OH,b}) \quad (B16) \quad \kappa_m = D_{Na,m}C_{Na,lm} + D_{H,m}C_{H,lm} \quad (B30)$$

Activity coefficients and activity-based pH [30,31]

To correct the reversible voltage for non-ideal electrolyte behavior, the model applies the Davies equation for ionic activity coefficients:

$$I = \frac{1}{2} \sum_i c_i z_i^2 \quad (B17)$$

$$\log_{10} \gamma_i = -A(T) z_i^2 \left(\frac{\sqrt{I}}{1 + \sqrt{I}} - 0.3I \right) \quad (B18)$$

$$a_i = \gamma_i c_i \quad (B19)$$

The activity-based pH values are therefore

$$pH_{a,act} = -\log_{10} (a_{H,a}) \quad (B20)$$

$$pH_{b,act} = -\log_{10} (a_{H,b}) \quad (B21)$$

Temperature-dependent transport properties

The diffusivities and ionic molar conductivities are corrected with temperature by Arrhenius-type expressions:

$$D_i(T) = D_{i,25} \left(\frac{T}{298.15} \right) \exp \left[1825 \left(\frac{1}{298.15} - \frac{1}{T} \right) \right] \quad (B22)$$

$$\lambda_i(T) = \lambda_{i,25} \left(\frac{T}{298.15} \right) \exp \left[1825 \left(\frac{1}{298.15} - \frac{1}{T} \right) \right] \quad (B23)$$

The electrolyte conductivity in each tank is then estimated from the ionic conductivities and corrected for ionic strength:

$$\sigma = \sum \lambda_i(T) c_i 1000 \max(1 - 0.23\sqrt{\min(I, 2)}, 0.3) \quad (B24)$$

Cation-exchange membrane model [26,27]

The membrane model includes sodium and proton transport through a cation-exchange membrane using a reduced Nernst-Planck formulation. Donnan partitioning is first imposed at each membrane interface:

$$C_{Na,0} = Q \frac{c_{Na,a}}{c_{Na,a} + c_{H,a}}, C_{H,0} = Q \frac{c_{H,a}}{c_{Na,a} + c_{H,a}} \quad (B25)$$

$$C_{Na,d} = Q \frac{c_{Na,b}}{c_{Na,b} + c_{H,b}}, C_{H,d} = Q \frac{c_{H,b}}{c_{Na,b} + c_{H,b}} \quad (B26)$$

The membrane diffusivities are reduced relative to their solution values by species-specific membrane-fraction factors. Diffusive and migrative fluxes are then written as

$$D_{i,m} = f_{i,m} D_i(T) \quad (B27)$$

$$J_i^{diff} = -D_{i,m} \frac{C_{i,d} - C_{i,0}}{d_m} \quad (B28)$$

$$C_{i,lm} = \frac{C_{i,d} - C_{i,0}}{\ln(C_{i,d}/C_{i,0})} \quad (B29)$$

$$-\frac{d\phi}{dx} = \left[\frac{j/F - (J_{Na}^{diff} + J_H^{diff})}{k_m 1000} \right] \frac{RT}{F} \quad (B31)$$

The migration fluxes and total membrane fluxes are then

$$J_i^{mig} = D_{i,m} C_{i,lm} 1000 \frac{F}{RT} \left(-\frac{d\phi}{dx} \right) \quad (B32)$$

$$J_{Na} = J_{Na}^{diff} + J_{Na}^{mig} \quad (B33)$$

$$J_H = J_H^{diff} + J_H^{mig} \quad (B34)$$

$$j = F(J_{Na} + J_H) \quad (B35)$$

The transport numbers and electro-osmotic water drag are computed as [28,29]

$$t_{Na} = \frac{J_{Na}}{J_{Na} + J_H}, t_H = \frac{J_H}{J_{Na} + J_H} \quad (B36)$$

$$J_{H_2O,drag} = n_{drag} \max(J_{Na} + J_H, 0) \quad (B37)$$

$$R_{Na} = J_{Na} A_{tot}, R_{H,back} = J_H A_{tot}, R_{H_2O,drag} = J_{H_2O,drag} A_{tot} \quad (B38)$$

The membrane potential is split into ohmic and Donnan contributions:

$$\eta_{mem,ohm} = \left(-\frac{d\phi}{dx} \right) d_m \quad (B39)$$

$$V_{Don,a} = -\frac{RT}{F} \ln \left(\frac{Q}{c_{Na,a} + c_{H,a}} \right) \quad (B40)$$

$$V_{Don,b} = -\frac{RT}{F} \ln \left(\frac{Q}{c_{Na,b} + c_{H,b}} \right) \quad (B41)$$

$$\eta_{mem} = V_{Don,b} - V_{Don,a} + \eta_{mem,ohm} \quad (B42)$$

CO₂ degassing and residual caustic neutralization

The release of CO₂ from the acid tank is modeled by a gas-liquid mass-transfer expression using a $k_L a$ term and Henry's-law solubility:

$$H_{CO_2}(T) = 3.4 \times 10^{-2} \exp \left[2400 \left(\frac{1}{T} - \frac{1}{298.15} \right) \right] \quad (B43)$$

$$r_{degas} = k_L a \max(c_{CO_2,aq} - H_{CO_2}(T) P_{CO_2}, 0) \quad (B44)$$

$$R_{degas} = r_{degas} V_a \quad (B45)$$

The model also includes a smoothed neutralization term for residual NaOH in the acid tank:

$$f_{res} = \frac{1}{1 + \exp \left[-\frac{c_{res} - 0.005}{0.001} \right]} \quad (B46)$$

$$R_{H,net} = \max(r_e - R_{H,back}, 0) \quad (B47)$$

$$R_{neutralize} = R_{H,net} f_{res} \quad (B48)$$

Electrode kinetics and voltage losses [32,33]

The exchange current densities at the anode and cathode are temperature-dependent. For both electrodes, the reference exchange current density is corrected by an Arrhenius relation, and the cathode value is additionally scaled by $c_{OH}^{1/2}$:

$$j_0(T) = j_{0,ref} \exp \left[-\frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right] \quad (B49)$$

$$j_{0,cath}(T) = j_{0,cath,ref} \exp \left[-\frac{E_{a,cath}}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right] c_{OH,b}^{1/2} \quad (B50)$$

The activation overpotentials are obtained by solving the Butler-Volmer relation for η :

$$|j| = j_0 \left[\exp \left(\alpha \frac{F\eta}{RT} \right) - \exp \left(-(1-\alpha) \frac{F\eta}{RT} \right) \right] \quad (B51)$$

The liquid-phase ohmic loss through the acid and base channels is

$$\eta_{ohm} = |j| \left(\frac{d_{acid}}{\sigma_a} + \frac{d_{base}}{\sigma_b} \right) \quad (B52)$$

Concentration overpotentials at both electrodes are included as

$$\eta_{con,an} = \frac{RT}{2F} \ln \left(1 - \frac{|j|}{j_{lim,an}} \right) \quad (B53)$$

$$\eta_{con,cath} = \frac{RT}{F} \ln \left(1 + \frac{|j| \delta_{BL}}{F D_{OH} c_{OH,b} 1000} \right) \quad (B54)$$

The reversible potential and total cell voltage are finally calculated from

$$E_{rev,act} = \frac{RT}{F} 2.303 (pH_{b,act} - pH_{a,act}) \quad (B55)$$

$$E_{cell} = E_{rev,act} + \eta_{an} + \eta_{cath} + \eta_{ohm} + |\eta_{mem}| + \eta_{con,an} + \eta_{con,cath} \quad (B56)$$

Thermal model [34, 35]

The model treats the entire electrochemical unit as a lumped thermal mass. Heat generation by irreversible voltage losses, the reversible entropic term, and external heat exchange with the ambient are included in the transient energy balance.

$$\eta_{irr} = \eta_{an} + \eta_{cath} + \eta_{ohm} + |\eta_{mem}| + \eta_{con,an} + \eta_{con,cath} \quad (B57)$$

$$C_{p,sol} = 4186 \max(1 - 0.07 \min(c, 5), 0.7) \quad (B58)$$

$$m_a = n_{H_2O,a} M_{H_2O} + n_{Na,a} M_{Na} + n_{DIC} M_{DIC} \quad (B59)$$

$$m_b = n_{H_2O,b} M_{H_2O} + n_{Na,b} M_{Na} + n_{OH,b} M_{OH} \quad (B60)$$

$$\frac{dT}{dt} = \frac{I_{tot} \eta_{irr} + I_{tot} T \frac{\Delta S}{F} - UA_{ext}(T - T_{amb})}{m_a C_{p,a} + m_b C_{p,b}} \quad (B61)$$

Performance equations [4,38]

After time integration, the code evaluates recovery, NaOH production, energy demand, balance closures, and second-law efficiency.

$$n_{CO_2,stoich} = c_{Na_2CO_3,feed} V_{acid,0} \quad (B62)$$

$$\text{Recovery} = \frac{n_{CO_2,g,final}}{n_{CO_2,stoich}} \times 100 \quad (B63)$$

$$E_{elec} = \int_0^{t_f} P(t) dt \quad (B64)$$

$$SEC = \frac{E_{elec}}{n_{CO_2,g,final}} \quad (B65)$$

$$\text{NaOH purity} = \frac{c_{NaOH,prod}}{c_{Na,b}} \times 100 \quad (B66)$$

$$\text{DIC balance} = \frac{n_{DIC,liq,final} + n_{CO_2,g,final}}{n_{DIC,0}} \times 100 \quad (B67)$$

$$\text{Na balance} = \frac{n_{Na,a,final} + n_{Na,b,final}}{n_{Na,a,0} + n_{Na,b,0}} \times 100 \quad (B68)$$

$$\eta_{2nd} = \frac{W_{min}}{SEC} \times 100, W_{min} = 2RT \cdot 2.303 \Delta pH \quad (B69)$$

The model also reports hydrogen coproduct generation from the consumed charge:

$$n_{H_2} = \frac{n_{e,consumed}}{2} \quad (B70)$$

Hydraulic and equipment-sizing calculations [36,37]

In addition to the electrochemical and species balances, the code includes simple engineering calculations for stack geometry, flow channels, pumps, tanks, gas separation, piping, and power supply sizing. These relations are used for equipment estimates rather than for dynamic state integration.

For each cell, the channel width and height are computed from the active area and a fixed aspect ratio, and the hydraulic relations are evaluated from

$$H = \sqrt{\frac{A_{cell}}{ar}}, W = ar H \quad (B71)$$

$$v = \frac{Q}{Wg} \quad (B72)$$

$$d_h = \frac{2Wg}{W + g} \quad (B73)$$

$$Re = \frac{\rho v d_h}{\mu} \quad (B74)$$

$$\Delta P_{ch} = f \frac{L}{d_h} \frac{\rho v^2}{2} \quad (B75)$$

The manifold pressure drop, pump power, and electrical power-supply demand are estimated from

$$\Delta P_{man} = 0.5 \rho v_m^2 \quad (B76)$$

$$P_{shaft} = \frac{Q \Delta P}{\eta_{pump}} \quad (B77)$$

$$P_{pump,elec} = \frac{P_{shaft}}{\eta_{motor}} \quad (B78)$$

$$P_{DC} = V_{stack} I_{stack} \quad (B79)$$

$$P_{AC} = \frac{P_{DC}}{\eta_{rect}} \quad (B80)$$

The stack length and internal hold-up volumes are computed from the number of cells and layer thicknesses:

$$L_{stack} = N_{cell} (t_{el} + g_a + d_m + g_b + t_{bp}) + 2t_{end} + t_{el} \quad (B81)$$

$$V_{acid,int} = N_{cell} A_{cell} g_a \quad (B82)$$

$$V_{base,int} = N_{cell} A_{cell} g_b \quad (B83)$$

For the storage tanks, the code assumes cylindrical vessels with 20% freeboard and a fixed $L/D = 1.5$ design rule:

$$V_{tank,total} = 1.2 V_{working} \quad (B84)$$

$$D_{tank} = \left(\frac{4V_{tank,total}}{\pi(L/D)} \right)^{1/3}, L_{tank} = (L/D) D_{tank} \quad (B85)$$

The CO₂ gas separator is sized from the average gas generation rate and a 30 s residence time:

$$\dot{V}_{CO_2} = \dot{n}_{CO_2} \frac{RT}{P} \quad (B86)$$

$$V_{sep} = \dot{V}_{CO_2} \tau_{res} \quad (B87)$$

Scale	CO ₂ (t/yr)	CAPEX (USD)	OPEX/yr (USD)	Rev/yr (USD)	Net/yr (USD)	LCOC (\$/t)	Payback (yr)	NPV (20yr) (USD)
1x	0.2	6,251	3,391	954	-436	7,976	>20	-12,952
5x	1.1	16,202	3,398	4,771	1,373	3,941	8.6	4,894
10x	2.1	24,442	5,299	9,543	4,243	3,054	4.9	40,732
20x	4.3	36,897	8,549	19,086	10,537	2,434	3.2	124,935
50x	10.7	63,658	16,847	47,714	30,867	1,876	2.0	410,428
100x	21.4	96,241	29,001	95,429	66,427	1,584	1.4	924,013
200x	42.7	145,594	51,013	190,857	139,844	1,365	1.0	2,002,260
500x	106.8	251,984	110,609	477,143	366,534	1,154	0.7	5,377,661

Table 1: Scaleup analysis – path to profitability

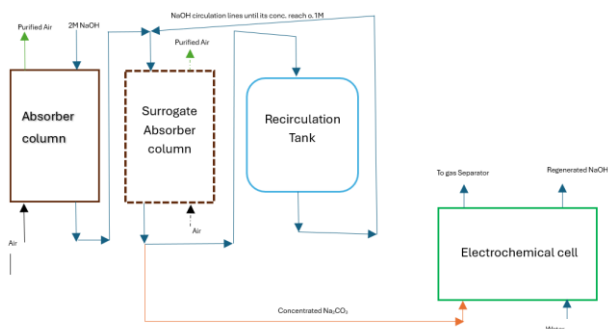


Figure 1: Block flow diagram of DAC